

Optimization: Critical Points & Lagrange Multipliers

ANSWER KEY



Finding critical points

- 1 Find the critical point(s) of $f(x, y) = x^2 + 3y^2 - 4x + 6y$.
 $f_x = 2x - 4 = 0 \Rightarrow x = 2$. $f_y = 6y + 6 = 0 \Rightarrow y = -1$. Critical point: $(2, -1)$.
- 2 Find the critical point(s) of $f(x, y) = xy - x^2 - y^2$.
 $f_x = y - 2x = 0 \Rightarrow y = 2x$. $f_y = x - 2y = 0 \Rightarrow x = 2y$. Substituting: $x = 2(2x) = 4x \Rightarrow x = 0$, so $y = 0$.
Critical point: $(0, 0)$.

The Hessian and the Second Derivative Test

- 3 Classify the critical point $(2, -1)$ of $f(x, y) = x^2 + 3y^2 - 4x + 6y$ using $D = f_{xx}f_{yy} - (f_{xy})^2$.
 $f_{xx} = 2$, $f_{yy} = 6$, $f_{xy} = 0$. $D = 2(6) - 0 = 12 > 0$ and $f_{xx} > 0$, so $(2, -1)$ is a local minimum.
- 4 Classify the critical point $(0, 0)$ of $f(x, y) = xy - x^2 - y^2$.
 $f_{xx} = -2$, $f_{yy} = -2$, $f_{xy} = 1$. $D = (-2)(-2) - 1^2 = 3 > 0$ and $f_{xx} < 0$, so $(0, 0)$ is a local maximum.
- 5 Classify the critical point of $f(x, y) = x^2 - y^2$.
 $f_x = 2x = 0$, $f_y = -2y = 0 \Rightarrow (0, 0)$. $f_{xx} = 2$, $f_{yy} = -2$, $f_{xy} = 0$. $D = 2(-2) - 0 = -4 < 0$, so $(0, 0)$ is a saddle point.

Lagrange multipliers: setting up the system

- 6 Set up the Lagrange multiplier equations to optimize $f(x, y) = xy$ subject to $g(x, y) = x + y - 10 = 0$.
 $\nabla f = \lambda \nabla g$: $\langle y, x \rangle = \lambda \langle 1, 1 \rangle \Rightarrow y = \lambda$, $x = \lambda \Rightarrow x = y$, together with the constraint $x + y = 10$.

Solving Lagrange multiplier problems

- 7 Maximize $f(x, y) = xy$ subject to $x + y = 10$.
From the setup above, $x = y$ and $x + y = 10 \Rightarrow x = y = 5$. Maximum value: $f(5, 5) = 25$.
- 8 Minimize $f(x, y) = x^2 + y^2$ subject to $x + y = 4$.
 $\nabla f = \lambda \nabla g$: $\langle 2x, 2y \rangle = \lambda \langle 1, 1 \rangle \Rightarrow x = y$. With $x + y = 4$: $x = y = 2$. Minimum value: $f(2, 2) = 8$.

A geometric Lagrange multiplier application

- 9 Find the point on the line $x + y = 6$ closest to the origin, using Lagrange multipliers on $f(x, y) = x^2 + y^2$.
 $\nabla f = \lambda \nabla g$: $\langle 2x, 2y \rangle = \lambda \langle 1, 1 \rangle \Rightarrow x = y$. With $x + y = 6$: $x = y = 3$. Closest point: $(3, 3)$, at distance $\sqrt{9+9} = 3\sqrt{2}$.

More critical point practice

- 10** Find the critical point(s) of $f(x, y) = x^2 + y^2 - 6x + 2y + 5$.

$$f_x = 2x - 6 = 0 \Rightarrow x = 3. f_y = 2y + 2 = 0 \Rightarrow y = -1. \text{ Critical point: } (3, -1).$$

- 11** Find the critical point(s) of $f(x, y) = x^3 - 3xy + y^3$.

$$f_x = 3x^2 - 3y = 0 \Rightarrow y = x^2. f_y = -3x + 3y^2 = 0 \Rightarrow x = y^2. \text{ Substituting:}$$

$$x = (x^2)^2 = x^4 \Rightarrow x^4 - x = 0 \Rightarrow x(x^3 - 1) = 0 \Rightarrow x = 0 \text{ or } x = 1. \text{ This gives } (0, 0) \text{ and } (1, 1).$$

More Hessian classification practice

- 12** Classify the critical point $(3, -1)$ of $f(x, y) = x^2 + y^2 - 6x + 2y + 5$.

$$f_{xx} = 2, f_{yy} = 2, f_{xy} = 0. D = 2(2) - 0 = 4 > 0 \text{ and } f_{xx} > 0, \text{ so } (3, -1) \text{ is a local minimum.}$$

- 13** Classify the critical points $(0, 0)$ and $(1, 1)$ of $f(x, y) = x^3 - 3xy + y^3$ found above.

$$f_{xx} = 6x, f_{yy} = 6y, f_{xy} = -3. \text{ At } (0, 0): D = 0(0) - 9 = -9 < 0, \text{ a saddle point. At } (1, 1):$$

$$D = 6(6) - 9 = 27 > 0 \text{ and } f_{xx} = 6 > 0, \text{ a local minimum.}$$

An applied unconstrained optimization problem

- 14** A firm's profit from producing x, y units of two goods is $P(x, y) = 100x + 120y - 2x^2 - 3y^2 - 2xy$. Find the production levels that maximize profit.

$$P_x = 100 - 4x - 2y = 0 \Rightarrow 2x + y = 50. P_y = 120 - 6y - 2x = 0 \Rightarrow x + 3y = 60. \text{ From the first, } y = 50 - 2x; \\ \text{substituting: } x + 3(50 - 2x) = 60 \Rightarrow -5x = -90 \Rightarrow x = 18, y = 14. \text{ Check: } P_{xx} = -4, P_{yy} = -6, P_{xy} = -2; \\ D = 24 - 4 = 20 > 0 \text{ and } P_{xx} < 0, \text{ confirming a maximum at } (18, 14).$$

Lagrange multipliers with a circular constraint

- 15** Set up the Lagrange multiplier equations to optimize $f(x, y) = x + y$ subject to $x^2 + y^2 = 8$.

$$\nabla f = \lambda \nabla g: \langle 1, 1 \rangle = \lambda \langle 2x, 2y \rangle \Rightarrow x = y, \text{ together with the constraint } x^2 + y^2 = 8.$$

- 16** Find the maximum and minimum values of $f(x, y) = x + y$ subject to $x^2 + y^2 = 8$.

$$\text{From } x = y \text{ and } x^2 + y^2 = 8: 2x^2 = 8 \Rightarrow x = \pm 2, \text{ giving points } (2, 2) \text{ and } (-2, -2). f(2, 2) = 4 \text{ (maximum),} \\ f(-2, -2) = -4 \text{ (minimum).}$$

An applied Lagrange multiplier problem

- 17** A farmer has 100 m of fencing to enclose a rectangular field along a straight river (no fence needed along the river side). If x is the side parallel to the river and y each perpendicular side, maximize the area $A = xy$ subject to $x + 2y = 100$.

$$\nabla A = \lambda \nabla g: \langle y, x \rangle = \lambda \langle 1, 2 \rangle \Rightarrow y = \lambda, x = 2\lambda \Rightarrow x = 2y. \text{ With } x + 2y = 100: 2y + 2y = 100 \Rightarrow y = 25, x = 50. \\ \text{Maximum area: } A = 50(25) = 1250 \text{ m}^2.$$

One more critical point and classification

- 18** Find and classify the critical point(s) of $f(x, y) = x^2 + xy + y^2$.

$f_x = 2x + y = 0$, $f_y = x + 2y = 0$. From the first, $y = -2x$; substituting:

$x + 2(-2x) = 0 \Rightarrow -3x = 0 \Rightarrow x = 0$, $y = 0$. $f_{xx} = 2$, $f_{yy} = 2$, $f_{xy} = 1$; $D = 4 - 1 = 3 > 0$ and $f_{xx} > 0$, so $(0, 0)$ is a local minimum.

A closing Lagrange multiplier problem

- 19** Minimize $f(x, y) = x^2 + y^2$ subject to $xy = 1$ (with $x, y > 0$).

$\nabla f = \lambda \nabla g$: $\langle 2x, 2y \rangle = \lambda \langle y, x \rangle \Rightarrow 2x = \lambda y$, $2y = \lambda x$. Dividing: $\frac{2x}{2y} = \frac{\lambda y}{\lambda x} \Rightarrow x^2 = y^2 \Rightarrow x = y$ (since both positive). With $xy = 1$: $x^2 = 1 \Rightarrow x = y = 1$. Minimum value: $f(1, 1) = 2$.

A final critical-point classification

- 20** Find and classify the critical point(s) of $f(x, y) = x^2y - 6y^2 - 3x^2$.

$f_x = 2xy - 6x = 2x(y - 3) = 0 \Rightarrow x = 0$ or $y = 3$. $f_y = x^2 - 12y = 0$. If $x = 0$: $y = 0$, giving $(0, 0)$. If $y = 3$: $x^2 = 36 \Rightarrow x = \pm 6$, giving $(6, 3)$ and $(-6, 3)$. At $(0, 0)$: $f_{xx} = 2y - 6 = -6$, $f_{yy} = -12$, $f_{xy} = 2x = 0$; $D = (-6)(-12) - 0 = 72 > 0$, $f_{xx} < 0$, a local maximum.

Global extrema on a closed region

- 21** Find the maximum and minimum of $f(x, y) = x^2 + y^2$ on the closed disk $x^2 + y^2 \leq 4$ (check both the interior critical point and the boundary).

Interior: $f_x = 2x = 0$, $f_y = 2y = 0 \Rightarrow (0, 0)$, with $f(0, 0) = 0$ (a minimum). On the boundary $x^2 + y^2 = 4$: $f = x^2 + y^2 = 4$ everywhere on the boundary. So the minimum value is 0 at $(0, 0)$, and the maximum value is 4, attained at every point on the boundary circle.

One last constrained-optimization setup

- 22** A box with a square base of side x and height y has volume $V = x^2y = 32$. Set up the Lagrange equations to minimize the surface area $S = 2x^2 + 4xy$ subject to this constraint.

$\nabla S = \lambda \nabla g$ for $g = x^2y - 32$: $\langle 4x + 4y, 4x \rangle = \lambda \langle 2xy, x^2 \rangle$, together with $x^2y = 32$.

- 23** Solve the box problem above: find x and y that minimize $S = 2x^2 + 4xy$ subject to $x^2y = 32$.

From $4x = \lambda x^2$ (second component, $x \neq 0$): $\lambda = \frac{4}{x}$. Substituting into the first:

$4x + 4y = \frac{4}{x}(2xy) = 8y \Rightarrow 4x = 4y \Rightarrow x = y$. With $x^2y = 32$ and $y = x$: $x^3 = 32 \Rightarrow x = 32^{1/3} = 2\sqrt[3]{4}$, so

$x = y = 2\sqrt[3]{4}$.

Second derivative test practice, continued

- 24** Find and classify the critical point(s) of $f(x, y) = 4 + x^3 + y^3 - 3xy$.

$$f_x = 3x^2 - 3y = 0 \Rightarrow y = x^2. f_y = 3y^2 - 3x = 0 \Rightarrow x = y^2. \text{ Substituting:}$$

$x = (x^2)^2 = x^4 \Rightarrow x^4 - x = 0 \Rightarrow x(x^3 - 1) = 0 \Rightarrow x = 0 \text{ or } x = 1$, giving $(0, 0)$ and $(1, 1)$. $f_{xx} = 6x$, $f_{yy} = 6y$, $f_{xy} = -3$. At $(0, 0)$: $D = 0 - 9 = -9 < 0$, a saddle point. At $(1, 1)$: $D = 36 - 9 = 27 > 0$, $f_{xx} = 6 > 0$, a local minimum.

A Lagrange multiplier problem in three variables

- 25** Set up the Lagrange equations to minimize $f(x, y, z) = x^2 + y^2 + z^2$ subject to $x + y + z = 6$.

$$\nabla f = \lambda \nabla g: \langle 2x, 2y, 2z \rangle = \lambda \langle 1, 1, 1 \rangle \Rightarrow x = y = z = \frac{\lambda}{2}, \text{ together with } x + y + z = 6.$$

- 26** Solve the problem above.

Since $x = y = z$ and $x + y + z = 6$: $3x = 6 \Rightarrow x = y = z = 2$. Minimum value: $f(2, 2, 2) = 4 + 4 + 4 = 12$.